

OJAS GRAMOPADHYE

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EDUCATION

New York University, New York Aug 2024 - May 2026
Master of Science, Computer Science GPA : 3.89/4
Courses : Artificial Intelligence, Deep Learning, Robot Perception, Big Data, Algorithms, ML Sys-Ops (Spring 2026)

Indian Institute of Technology Bombay, India May 2019 - Jul 2023
Bachelor of Technology, Computer Science and Engineering GPA : 7.74/10
Courses : AI & ML, Speech Recognition, Image Processing, Database Systems, Data Structures, Software Systems

KEY PROJECTS

Multi-subject alignment : Neural Speech Representation Learning Sept 2025 - Ongoing
Graduate Research | NYU Video Lab **New York University**

- Developed a multi-subject neural speech decoding framework that learns subject-invariant ECoG representations via contrastive alignment and adapter-based modeling to improve cross-subject generalization.
- Implemented wav2vec2-style self-supervised pretraining for neural signals, using masked encoder modeling, Gumbel quantization, and Transformer context modeling to enhance downstream phoneme classification.

Erica : AI Tutor with GraphRAG Sept 2025 - Dec 2025
Course Project | Artificial Intelligence **New York University**

- Developed a Dockerized LLM tutor agent that ingests multi-format course content, enriches it with LLM metadata, stores corpus artifacts in MongoDB, and pipelines them into GraphRAG for grounded, concept-aware retrieval
- Delivered structured, prerequisite-aware explanations by integrating semantic search, a typed knowledge graph, and Streamlit UI with subgraph visualizations, enabling the agent to answer queries with references

Modulus : Agentic AI Assistant Jun 2025 - Aug 2025
Independent Project **New York University**

- Implemented an agentic LLM system using MCP and LangGraph, constructing explicit execution graphs to support multi-turn, multi-threaded conversations, dynamic tool routing, and stateful LLM execution with streaming outputs
- Built a persistent agent memory architecture using ChromaDB and MongoDB, enabling vector-based retrieval for short- and long-horizon context, and integrating the system through a scalable FastAPI backend with a React frontend

WORK EXPERIENCE

Detection and Mitigation of Hallucination in Generative Models | EMNLP '24 May 2023 - May 2024
Research Assistant **IBM Research | IIT Bombay**

- Proposed a forward-backward clinical reasoning framework (CLINICR) for medical QA, combining differential diagnosis with autoregressive LLM reasoning to reduce hallucinations over eliminative CoT baselines.
- Fine-tuned LLaMA-2 (7B & 70B) with reward-based verification on a new Subjective Medical QA benchmark, achieving 90% expert-validated diagnostic accuracy. (EMNLP Findings 2024) arXiv: arxiv.org/abs/2403.04890

Japanese OCR Pipeline May 2022 - Jul 2022
Data Science Intern **Daikin**

- Delivered a full OCR pipeline for 3,000+ Japanese documents by building document-cleaning and image-normalization scripts, achieving 0.91 character-level accuracy with a MobileNetV3 + Differentiable Binarization detector
- Wrote preprocessing and evaluation pipeline to streamline dataset preparation and measure OCR performance on proprietary scanned documents

Locust Occurrence Modeling and Prediction May 2021 - Jul 2021
Data Science Associate **Dtime.ai**

- Supported an early-warning system for East African farmers by processing 40+ years geospatial vegetation and weather data from East African region into structured 10x10 square-km grids and handling class imbalance with SMOTE
- Improved predicted locust attack precision by 10% over baselines by performing pixel-level classification and modeling spatiotemporal patterns with a ConvLSTM U-Net model

TECHNICAL SKILLS

Machine Learning: LLMs, RAG systems, AI agents, NLP, deep learning, contrastive learning, representation learning, LoRA fine-tuning
Frameworks & Libraries: PyTorch, TensorFlow, HuggingFace, PyTorch Lightning, LangGraph, LangChain, SentenceTransformers, scikit-learn
Data & Systems: Apache Kafka, Apache Spark, Dask, MongoDB, PostgreSQL, ChromaDB, MLflow, Docker, Linux, Bash
Software Engineering: Python, C++, JavaScript, TypeScript, FastAPI, React, Streamlit, d3.js, Git, REST APIs